Local Consumer Complaint Classification: Zero-Shot, Lightweight Adaptation and Label Budgets

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Abstract

Local classification of consumer financial complaints must balance accuracy, label availability, inference latency and GPU memory. We study these engineering constraints on a five-category Consumer Financial Protection Bureau task using an archived zero-shot benchmark of eleven open-weight models, supervised classifiers and controlled lightweight adaptation. The preserved protocol comprises balanced and reduced-imbalance conditions, three test subsets and paired 250/500-character narratives. We adapt Gemma 4 12B, Qwen3 14B and Phi-4 14B with QLoRA using 500 balanced training records and 2,000 separate validation records. Each adapter is compared with its own base model under the same Hugging Face revision, NF4 quantisation, chat template and thinking-disabled configuration. On the pooled 3,000 balanced 500-character records, weighted F1 increases from 0.7520 to 0.7803, 0.7099 to 0.7891 and 0.6883 to 0.7845, respectively. Paired bootstrap intervals are positive in all twelve model–condition–length comparisons, and no adapted prediction is outside the taxonomy. Training takes approximately 35–41 minutes with 7.77–10.04 GiB peak allocated memory on a 16 GB RTX 5060 Ti. Warm serial mean latency ranges from 0.435 to 0.647 s for adapted models; adaptation does not uniformly reduce latency. TF-IDF/SVM reaches F1 0.7546 with 500 training records and 0.8291 with 20,000, while three DistilBERT fits average 0.8244 at 20,000. These unequal label budgets support conditional deployment choices rather than a universal model ranking. Historical API comparisons remain contextual, and the single adaptation training seed limits generalisability.

Keywords: consumer complaint classification, large language models, QLoRA, label budgets, CFPB, local deployment

1 Introduction

The Consumer Financial Protection Bureau (CFPB) publishes complaint narratives that can support regulatory oversight and financial-service routing. Misclassification may delay a complaint or send it to the wrong department, motivating both accuracy and operational reliability [1–4]. Banks and regulators may also need to keep new complaint text on local infrastructure rather than transmit it to third-party APIs.

Supervised text classifiers require task-specific training labels; zero-shot instruction-following models instead receive the taxonomy in a prompt [5–7]. This reduces adaptation effort when taxonomies change, but introduces engineering costs in generation latency, output parsing, and hardware. A model that invents an unsupported label cannot be routed safely through a system that assumes a fixed category set.

The closest published comparison, Roumeliotis et al. [8], evaluates 14 commercial model APIs on the five-category CFPB task. Its best result is o1 at accuracy 0.777 and weighted F1 0.7764, with a reported historical API charge of $24.24 per thousand predictions. We examine whether local open-weight models provide competitive zero-shot performance, and how that choice changes when labelled training data are available. We evaluate balanced and reduced-imbalance samples, paired narrative-length variants, invalid outputs, and inference latency. Exact commercial test identifiers are unavailable, so cross-study comparisons remain protocol-matched rather than instance-identical.

We additionally ask whether 500 labelled complaints suffice to improve three representative LLMs within a 16 GB GPU budget, and whether those gains justify their training and serving costs relative to direct classifiers. The supervised experiments share a separate 2,000-record validation set; training-label budgets exclude these additional labels. QLoRA enables this test through low-rank updates to frozen quantised weights [9, 10]. The experiment preserves every existing test file and selects adapters using validation loss, keeping test outcomes outside model selection.

Our contributions are:

1. We benchmark eleven open-source, open-weight LLMs — `gemma2:9b`, `gemma4:12b`, `mistral:7b`, `mistral-nemo:12b`, `llama3.1:8b`, `phi4:14b`, `deepseek-r1:8b`, `deepseek-r1:14b`, `qwen3:8b`, `qwen3:14b`, and `gpt-oss:20b` — on the same five-category CFPB zero-shot task, at **zero API cost**. This expanded model set includes matched 8B/14B variants for Qwen3 and DeepSeek-R1, enabling within-line scale comparisons; other family comparisons also change model generation.
2. We measure lightweight adaptation of Gemma 4 12B, Qwen3 14B and Phi-4 14B using 500 balanced labels, with paired adapters-on/off inference at identical model revisions and runtime settings, serial latency, training time and peak allocated GPU memory.
3. We conduct a second experiment using a reduced-imbalance class distribution, evaluating whether performance patterns observed under artificial balance hold in a more realistic scenario.
4. We provide a direct quantitative comparison with the paid-API results of Roumeliotis et al. [8], showing that our best open-weight model has a numerically higher score than GPT-4o under a matched evaluation protocol and remains competitive with more expensive API-based alternatives.

5. We analyse off-label prediction behaviour — instances where models invent labels outside the five prescribed categories — as a proxy for instruction-following reliability.
6. We add supervised classical and DistilBERT baselines using training/validation data separated from every saved test subset, together with paired bootstrap uncertainty, multiplicity-adjusted accuracy tests and equal-prior reweighting. The classical learning curve separates 500/2,000/10,000/20,000-label budgets; LLM adaptation is evaluated at 500 labels and one training seed.

The remainder of this paper is structured as follows. Section 2 surveys the related literature. Section 3 describes the dataset and preprocessing pipeline. Section 4 details the experimental setup. Section 5 presents results for both experiments. Section 6 discusses the findings. Section 7 concludes with directions for future work.

2 Related Work

Classical complaint classifiers use engineered text features and supervised learners. Word embeddings and contextual transformers extend these representations [5, 11]. Prior CFPB work applies topic models and transformer classifiers, while complaint prioritisation and explainability studies address operational deployment [1, 2, 12–14].

Instruction-following LLMs enable zero-shot classification alongside summarisation and response generation [6, 15, 16]. Related tourism, e-commerce, and customer-service studies compare generative and supervised models, but their taxonomies and data splits differ from this task [17–19]. Recent analyses of LLM-assisted complaint writing provide additional motivation for robustness studies on newer corpora [20, 21]; our 2020–2021 data do not directly test that shift.

Reasoning-oriented models generate intermediate text before an answer [22–25]. Roumeliotis et al. [8] provide our closest five-category zero-shot reference. Bulgakov and Turchin [26] instead study binary monetary-relief prediction with a distilled local model. Güllü et al. [27] report 89.99 % accuracy for supervised LLaMA 3.1 8B; their different split and training procedure make this context rather than a target or directly comparable ceiling.

Low-rank adaptation freezes base parameters and learns small task-specific updates [9]; QLoRA combines these updates with NF4, double quantisation and paged optimisation [10]. We use these established methods to test the practical value of limited complaint labels on consumer hardware. Our contribution is an empirical local deployment comparison across label budgets and output-reliability requirements, with preserved test sets for supervised baselines. We do not claim a new LLM architecture. The extended background discussion is supplied in the supplementary material.

3 Dataset and Preprocessing

3.1 Data Source

We use the CFPB Consumer Complaints dataset originally published by Tiwari [28] on Kaggle, which is a processed subset of the public CFPB Consumer Complaint Database.

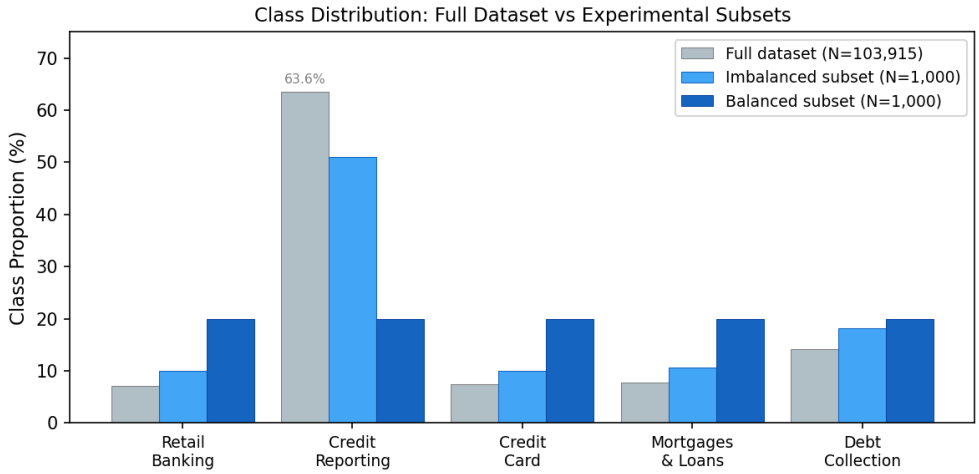


Fig. 1 Class proportions (%) in the full post-filter dataset (N=103,915), the imbalanced experimental subset (N=1,000), and the balanced subset (N=1,000). In the imbalanced condition, `credit_reporting` is still the dominant class at 51.1 %, confirming that balanced evaluation is essential to avoid majority-class inflation.

The dataset covers complaints submitted between March 2020 and March 2021 and contains three columns: an unnamed index, `product` (the category label), and `narrative` (the free-text complaint). The raw file comprises 162,421 records consolidated to five target product categories: `retail_banking`, `credit_reporting`, `credit_card`, `mortgages_and_loans`, and `debt_collection`. This is the same dataset and label set used by Roumeliotis et al. [8], ensuring direct comparability.

3.2 Class Distribution

The original data are heavily imbalanced. After removing records with empty `product` or `narrative` fields (10 rows removed, 162,411 remaining) and discarding narratives exceeding 500 characters (resulting in 103,915 usable records), the class distribution shifts to: `credit_reporting` 63.6 %, `debt_collection` 14.1 %, `mortgages_and_loans` 7.7 %, `credit_card` 7.5 %, and `retail_banking` 7.1 %. Before the length filter, the same five-category corpus contains 162,411 records with `credit_reporting` accounting for 56.1 % of the data. Fig. 1 illustrates how this skew propagates into the experimental subsets: the imbalanced condition retains the dominant `credit_reporting` class at 51.1 %, while the balanced condition enforces equal class sizes of 200 samples each. At N=1,000, a model that defaults to the majority class in the imbalanced setting would achieve 51.1 % accuracy with zero language understanding — motivating the need to report both conditions side by side.

3.3 Preprocessing Pipeline

The preprocessing steps, implemented in Python using pandas, are applied in the following order to every record:

1. Drop rows with empty `product` or `narrative`.
2. Retain only records belonging to the five target categories.
3. Remove narratives longer than 500 characters, reducing processing time and GPU memory pressure while avoiding truncation artifacts during inference.
4. Remove exact duplicate narratives before normalisation.
5. Standardise narrative text: convert to lowercase, remove special characters, and normalise whitespace.

3.4 Experimental Subsets

Two experimental conditions are defined, each evaluated under a multi-seed, multi-size design to ensure statistically reliable estimates.

Experiment A (Balanced): 200 records are sampled uniformly at random from each of the five categories (N=1,000 total) using stratified undersampling. This follows the evaluation protocol of Roumeliotis et al. [8]; because the original sample identifiers are not publicly released, the comparison should be interpreted as protocol-matched rather than instance-identical.

Experiment B (Reduced imbalance): The saved subsets have a less skewed distribution than the post-filter corpus, with at least 100 samples per class. They do not sample the unaltered natural distribution. Each contains N=1,000 with `credit_reporting` 51.1 %, `debt_collection` 18.2 %, `mortgages_and_loans` 10.7 %, `credit_card` 10.0 %, and `retail_banking` 10.0 %. The dominant `credit_reporting` class means a naive majority baseline would reach only 51.1 % accuracy with no language understanding. This second experiment, absent from the closest antecedent benchmark, assesses how class imbalance affects both accuracy estimates and per-class performance patterns.

Multi-seed and narrative-length design: Each condition is replicated with three subset identifiers (1, 2, and 3) and at two narrative character-length limits (250 and 500 characters), yielding 12 datasets per model and 132 inference runs in total (11 models $\times$ 3 seeds $\times$ 2 conditions $\times$ 2 length limits). The 250-character subsets contain exactly the same records in the same order as their 500-character counterparts, with narratives truncated to 250 characters. Balanced and reduced-imbalance subsets share records and are therefore not independent replications. The archived Ollama result tables use the 500-character variant and mean $\pm$ standard deviation across the three test subsets. New HF adaptation comparisons instead pool 3,000 predictions per mode and condition, with paired cluster-bootstrap intervals. The three test identifiers are distinct from the single LLM adaptation training seed. The 250-character runs provide a narrative-length sensitivity check discussed in Section 5.5. This design addresses the single-seed limitation common in prior zero-shot LLM benchmarks and allows us to distinguish robust ranking patterns from seed-specific artefacts.

The two length variants compare truncated and retained text from the same short-narrative population. The narrative-length histogram is provided in Supplementary Fig. S1.

4 Experimental Setup

4.1 Infrastructure

The new supervised and paired HF adaptation experiments run on an Ubuntu workstation with an Intel Core i7-14700 processor, 32 GB RAM, and an NVIDIA GeForce RTX 5060 Ti with 16 GB VRAM. The adaptation environment uses Python 3.12, PyTorch 2.14.1, Transformers 5.18.0, PEFT 0.21.2 and bitsandbytes 0.50.2. Model revisions, dataset and adapter SHA-256 hashes, package versions and run statuses are saved. The HF evaluation uses cached weights in offline mode; complaint narratives are not sent to a third-party model service.

The archived zero-shot benchmark uses Ollama, a self-contained serving runtime that manages model loading, KV-cache management, and the local REST API. Its Python inference script communicates through a local REST endpoint. Historical artifact digests, thinking modes and execution logs are incomplete; the saved results do not establish every historical hardware or serving configuration. New HF timing is consequently reported separately from archived Ollama timing.

4.2 Models

Table 1 lists all eleven models. The original grouping treated five builds as reasoning models and six as standard models. However, upstream Gemma 4 supports configurable thinking [29]; the exact thinking mode of the saved `gemma4:12b` runs was not recorded and its grouping remains unverified. Only Qwen3 and DeepSeek-R1 provide within-line 8B/14B pairs; other comparisons mix parameter count, generation, and training recipe. The builds use compressed GGUF weights. Current local metadata identifies `qwen3:8b` as Q4_K_M and `gpt-oss:20b` as MXFP4, so a uniform Q4_K_M claim is inappropriate. Current metadata does not establish the identity of every historical artifact; missing digests remain a reproducibility limitation. The largest model, `gpt-oss:20b`, requires 13.0 GB VRAM and fits within the 16 GB local GPU memory budget. The evaluated model families build on recent open-weight model lines, including LLaMA [30], Mistral [31], Gemma [32], Phi [33], Qwen [25], and DeepSeek [23, 24]. Model names in Table 1 are the exact Ollama tags used in the experiments. The family column records the upstream lineage or registry metadata associated with each tag and should not be interpreted as equivalence to a proprietary API model with a similar provider name. For this archived zero-shot benchmark, artifacts were obtained through `ollama pull <tag>` without task-specific fine-tuning. The new adaptation experiment uses separately pinned HF weights, as detailed below. `gpt-oss:20b` is an open-weight reasoning model released by OpenAI and distributed through the Ollama registry; it requires no API key and its weights are publicly accessible via `ollama pull gpt-oss:20b`. The "OpenAI" entry in the Family column reflects the model's originating developer, not access through any proprietary API or commercial service.

Table 1 Open-weight model builds evaluated. Historical artifact sizes are reported; peak runtime VRAM was not recorded.

Model	Family	Type	Params	Size
`gemma2:9b`	Google	Standard	9B	5.4 GB
`gemma4:12b`	Google	Unverified	12B	7.6 GB
`mistral:7b`	Mistral	Standard	7B	4.4 GB
`mistral-nemo:12b`	Mistral	Standard	12B	7.1 GB
`llama3.1:8b`	Meta	Standard	8B	4.9 GB
`phi4:14b`	Microsoft	Standard	14B	9.1 GB
`deepseek-r1:8b`	DeepSeek	Reasoning	8B	5.2 GB
`deepseek-r1:14b`	DeepSeek	Reasoning	14B	9.0 GB
`qwen3:8b`	Alibaba	Reasoning	8B	5.2 GB
`qwen3:14b`	Alibaba	Reasoning	14B	9.3 GB
`gpt-oss:20b`	OpenAI	Reasoning	20B	13.0 GB

4.3 Prompt Design

A single, model-agnostic prompt is used for all models and all experiments. The prompt follows the five-label JSON-output template used by Roumeliotis et al. [8] to ensure comparability:

> *You are an AI assistant specializing in consumer complaint classification. Your task is to analyze a consumer complaint and classify it into the most appropriate category from the predefined list: ["retail_banking", "credit_reporting", "credit_card", "mortgages_and_loans", "debt_collection"]. Provide your final classification in the following JSON format without explanations:* `{"product": "chosen_category_name"}`*. Complaint: [NARRATIVE]*

For the archived Ollama runs, temperature is set to 0 to reduce sampling variability and improve reproducibility. No explicit `top_p`, `max_tokens`, stop-sequence, or model-specific thinking-mode flags are set; Ollama defaults are used for all other decoding parameters. Although the prompt asks for JSON without explanations, some reasoning-oriented builds may still emit intermediate text before the final answer. The inference script therefore extracts the final JSON object from the response by searching for the last valid `{"product": "..."}` occurrence using a regular expression, accounting for models that prepend reasoning traces before the answer.

The inference script was implemented using only Python standard-library modules (`csv`, `json`, `urllib`, `time`, `argparse`), eliminating the need for any `pip`-installed dependencies on the compute node. Each request is retried once on HTTP failure before being recorded as an error; errors are included in the final count. Saved responses are limited to 2,000 characters and newlines are replaced with spaces. The legacy parser also falls back to a category mentioned anywhere in the response when no JSON match is found, which can accept explanatory text as a classification. The CSVs omit separate thinking text, termination reasons, and token counts, limiting post-hoc auditing. The full experiment was orchestrated as local batch runs totalling 132 tasks (11 models $\times$ 12 datasets), with runs scheduled sequentially or in controlled parallel groups according to GPU memory availability.

4.4 Evaluation Metrics

Following the closest antecedent benchmark, we compute four primary metrics over all N=1,000 predictions per run: *exact-match accuracy*, *weighted precision*, *weighted recall*, and *weighted F1-score*. We additionally report *macro-F1*, which weights all classes equally and is especially informative for the imbalanced experiment. Weighted variants weight each class by its sample size, making them appropriate for comparison with the reference paper but potentially optimistic under class imbalance. We additionally record the *mean latency* (seconds per prediction, wall-clock time) and the number of *off-label* predictions — responses where the model returned a category name not in the prescribed five, or failed to return valid JSON entirely. Off-label predictions are counted as errors. The API cost for all experiments is $0. Compute time is supplied by the local workstation rather than a commercial LLM API.

4.5 Controlled Lightweight LLM Adaptation

We select Gemma 4 12B, Qwen3 14B and Phi-4 14B to represent the leading archived zero-shot model, a configurable-thinking model and a fast generative classifier. The HF artifacts are `google/gemma-4-12B-it`, `Qwen/Qwen3-14B` and `microsoft/phi-4`. Their exact revisions are listed in the supplementary material. These are new HF experiments; archived Ollama predictions are not their adaptation controls.

Each model uses the same balanced 500-record training subset (100 per class) and the common 2,000-record validation set (400 per class), prepared with the test-exclusion procedure in Section 5.9. Reported training budgets exclude the additional validation labels. We use LoRA [9] with rank 8, scaling factor 16 and dropout 0.05, applied to attention and feed-forward projections. Gemma and Qwen use separate query/key/value, output, gate/up/down projections; Phi uses fused QKV and gate/up projections plus output/down projections. QLoRA [10] freezes NF4 double-quantised linear weights with BF16 compute. Unquantised matrices remain BF16 and one-dimensional normalisation parameters FP32. Only adapter weights are updated.

Training uses seed 1, three epochs, batch size 1, gradient accumulation 16, learning rate 2×10^{-4}, paged 8-bit AdamW, cosine decay, three warmup steps, gradient checkpointing and a 512-token cap. Loss is restricted to the assistant's target JSON and end-of-turn token; the prompt is masked. All training and validation sequences fit without truncating a target. Validation is evaluated after each epoch; minimum completion-only validation loss selects the checkpoint. All runs complete 96 optimiser steps, with Gemma's selected checkpoint at epoch 2 and Qwen/Phi at epoch 3. No test score selects a checkpoint or hyperparameter.

Paired inference loads one NF4 PEFT model and enables/disables its unmerged adapter. Both modes retain identical frozen weights, normalisation, chat template and precision. The same five-category prompt is rendered through each pinned model's chat template with thinking disabled. Greedy decoding uses a 64-new-token cap and the model's end-of-turn token, with independent KV caches for every call. Inputs are left-padded with attention masks. Maximum batch size is 32; Phi falls back to 16 after a long-input memory failure. Both modes of a pair use the same actual batch size. Full raw responses, token counts, output-limit flags, source hashes and original row indices

are saved. A JSON decoder accepts the last explicit string-valued `product` field; parse failures and out-of-taxonomy labels remain errors.

The full design yields 72,000 predictions: three models, two modes and twelve preserved 1,000-record files. Metrics pool the three test subsets within each condition/length. We compute 5,000 paired bootstrap resamples using identical normalised-text-cluster weights for both modes, stratified by class when cluster labels are consistent. Exact two-sided McNemar tests compare accuracy on unique source inputs, retaining the first subset/index for duplicates; conflicting-label inputs would be excluded only from this test. Holm correction covers all twelve model–condition–length comparisons. No independence between overlapping conditions or length variants is assumed. Confidence intervals describe these test outputs, not variation over training seeds.

Training/validation runtime excludes initial weight loading; peak allocated CUDA memory includes the loaded model and training operations. Serving latency is measured separately on the first 20 records per class in the first balanced 500-character subset (100 per mode/model), sorted back to their original indices, after a warmup call in each mode. Batch-1 timing includes prompt rendering, tokenisation and synchronised CUDA generation, excluding output decoding and network overhead. Full-test batch wall time divided by batch size is saved as amortised time and is not interpreted as serial latency.

5 Results

5.1 Experiment A: Balanced Evaluation

Table 2 presents the full results for Experiment A, reported as means over three test subsets. The `gemma4:12b` model achieves the highest accuracy (0.774 ± 0.017) and weighted F1 (0.774 ± 0.016), numerically exceeding the larger `gpt-oss:20b` (F1 0.749 ± 0.019) and ranking first among all eleven saved model runs. Its unrecorded thinking mode prevents using this result as evidence that reasoning is unnecessary. Second place goes to `qwen3:14b` (F1 0.750 ± 0.011), a reasoning model, followed by `gpt-oss:20b` in third. `deepseek-r1:14b` (F1 0.724 ± 0.007) and `qwen3:8b` (F1 0.717 ± 0.002) complete the top five. The standard `phi4:14b` (F1 0.706 ± 0.011) ranks sixth. Notably, `deepseek-r1:8b` (rank 7, F1 0.683 ± 0.012) only narrowly exceeds the standard `gemma2:9b` (rank 8, F1 0.676 ± 0.009), confirming that 8B reasoning models do not automatically dominate similarly sized standard models. The weakest model is `mistral-nemo:12b` (F1 0.540 ± 0.025, rank 11), which despite being a 12B model scores below both `mistral:7b` (F1 0.587) and `llama3.1:8b` (F1 0.587), suggesting that the Mistral Nemo architecture is not well-suited to strict JSON output tasks in the zero-shot setting. The low standard deviations across most models — notably `qwen3:8b` (± 0.002) and `mistral:7b` (± 0.004) — show limited score variation across these test subsets; small standard deviations alone do not establish every rank separation.

The `OL` column in Table 2 is particularly informative. The three zero-off-label models among the top-five (`qwen3:14b`, `gpt-oss:20b`, and `qwen3:8b`) are all reasoning models. The two models with severe format-following failures are standard models: `llama3.1:8b` produces 67 off-label predictions and `mistral:7b` produces 59. Importantly, `gemma4:12b` (rank 1) produces only 3.7 invalid predictions on average. Its saved

Table 2 Experiment A Results — Balanced (N = 1,000, mean of 3 seeds; std in parentheses)

Model	T	Acc.	F1w	F1m	Lat.(s)	OL
gemma4:12b	U	0.774(±.017)	0.774(±.016)	0.774	17.11	3.7
qwen3:14b	R	0.751(±.011)	0.750(±.011)	0.750	5.73	0.0
gpt-oss:20b	R	0.748(±.020)	0.749(±.019)	0.749	4.13	0.0
deepseek-r1:14b	R	0.724(±.006)	0.724(±.007)	0.724	9.33	1.3
qwen3:8b	R	0.717(±.003)	0.717(±.002)	0.717	5.19	0.0
phi4:14b	S	0.705(±.012)	0.706(±.011)	0.706	0.54	1.0
deepseek-r1:8b	R	0.684(±.013)	0.683(±.012)	0.683	5.14	2.0
gemma2:9b	S	0.680(±.006)	0.676(±.009)	0.676	0.67	0.3
mistral:7b	S	0.593(±.002)	0.587(±.004)	0.587	0.23	58.7
llama3.1:8b	S	0.578(±.005)	0.587(±.003)	0.587	2.97	66.7
mistral-nemo:12b	S	0.559(±.019)	0.540(±.025)	0.540	0.59	0.7

T: R=Reasoning, S=Standard, U=Unverified mode. OL=Invalid output mean. Lat.=Mean latency.

failures are empty responses recorded as parse errors, rather than valid out-of-taxonomy labels. Their cause cannot be determined from the retained response field alone. At the same time, `phi4:14b` is notable among standard models: it reaches F1 0.706 with only one off-label prediction and a mean latency of 0.54 s, making it the strongest low-latency alternative.

We perform post-hoc exact McNemar tests (binomial, one-sided) on paired correctness outcomes for each of the three seeds independently, applying a Bonferroni-adjusted threshold of α=0.0125 (four comparisons anchored on `gemma4:12b`). The advantage of `gemma4:12b` over `deepseek-r1:14b` and `qwen3:8b` is significant in all three seeds (p<0.001 each), providing robust statistical evidence for those rank separations. Comparisons with `qwen3:14b` and `gpt-oss:20b` reach significance in only one seed each. Non-significance in other subsets does not establish equality, and McNemar tests accuracy rather than F1. The two lowest-performing models (`llama3.1:8b`, `mistral-nemo:12b`) are clearly separated from all others by a wide margin.

For Experiment A, macro-F1 is numerically identical to weighted F1 because all five classes have equal support. Per-class inspection reveals that even the best model is uneven across labels: `gemma4:12b` reaches F1 0.825 on `retail_banking` and 0.820 on `mortgages_and_loans` but drops to 0.718 on `debt_collection` and 0.743 on `credit_reporting`. This motivates the macro-F1 reporting in Experiment B, where minority-class behaviour can otherwise be hidden by the majority class. Detailed per-class F1 for the leading models are reported in Table 5.

Supplementary Figs. S2–S5 show the balanced F1 distribution and confusion matrices for gpt-oss, Phi-4, and DeepSeek-R1 14B. Mortgage/loan and credit-card confusions remain visible even for models with almost no invalid outputs. Llama 3.1's larger invalid-output count primarily reflects refusals rather than invented category names.

Table 3 Experiment B Results — Imbalanced (N = 1,000, mean of 3 seeds; std in parentheses)

Model	T	Acc.	F1w	F1m	Lat.(s)	OL
gpt-oss:20b	R	0.782(±.017)	0.783(±.016)	0.740	4.01	0.0
gemma4:12b	U	0.776(±.015)	0.779(±.014)	0.741	10.61	2.7
qwen3:14b	R	0.772(±.012)	0.772(±.013)	0.730	5.53	0.0
qwen3:8b	R	0.766(±.008)	0.762(±.007)	0.719	6.97	0.3
phi4:14b	S	0.753(±.004)	0.754(±.002)	0.696	0.58	0.0
deepseek-r1:14b	R	0.746(±.017)	0.748(±.015)	0.700	7.32	0.3
deepseek-r1:8b	R	0.738(±.012)	0.731(±.011)	0.677	4.46	2.3
gemma2:9b	S	0.720(±.012)	0.719(±.014)	0.656	0.67	0.3
mistral-nemo:12b	S	0.653(±.011)	0.641(±.013)	0.538	0.59	0.7
mistral:7b	S	0.649(±.010)	0.661(±.010)	0.566	0.23	42.3
llama3.1:8b	S	0.632(±.006)	0.647(±.012)	0.564	0.56	57.3

T: R=Reasoning, S=Standard, U=Unverified mode. OL=Invalid output mean. Lat.=Mean latency.

5.2 Experiment B: Imbalanced Evaluation

Table 3 presents results for Experiment B, again as means over three seeds. All eleven models show numerically higher accuracy and F1 compared to Experiment A. `gpt-oss:20b` regains the top position under imbalanced conditions, with mean accuracy 0.782 ± 0.017 and F1 0.783 ± 0.016. `gemma4:12b` follows closely in second place (Acc 0.776 ± 0.015, F1 0.779 ± 0.014), and `qwen3:14b` ranks third (F1 0.772 ± 0.013). The top five models (`gpt-oss:20b`, `gemma4:12b`, `qwen3:14b`, `qwen3:8b`, `phi4:14b`) all achieve weighted F1 ≥ 0.754. The imbalanced setting changes the absolute score level more than the identity of the leading models. However, the gains for `llama3.1:8b` and `mistral:7b` should not be interpreted as genuine robustness improvements, because both models still produce many off-label outputs and benefit from the dominant `credit_reporting` class.

The relative model ranking is broadly preserved compared to Experiment A, with one notable reordering: `gpt-oss:20b` reclaims the top position while `gemma4:12b` drops to second. The macro-F1 column makes the imbalance effect visible: `gpt-oss:20b` has weighted F1 0.783 but macro-F1 0.740, while `llama3.1:8b` has weighted F1 0.647 but macro-F1 0.564. The gap shows that weighted metrics substantially overstate minority-class performance in the imbalanced setting. The F1-score distribution mirrors Experiment A in its broad ordering (Fig. 2 right bars), with `gpt-oss:20b`, `gemma4:12b`, and `qwen3:14b` forming the leading group.

5.3 Cross-Experiment Comparison

Fig. 2 provides a side-by-side view of accuracy across both experiments. All models improve numerically from Experiment A to Experiment B. `mistral-nemo:12b` has the largest gain (Δacc = +0.094), despite few invalid outputs. `mistral:7b` (+0.055) and `llama3.1:8b` (+0.054) also improve, but these shifts cannot be attributed solely to invalid outputs. Equal-prior reweighting separates the arithmetic effect of class proportions from class-conditional performance differences. The conditions use partly

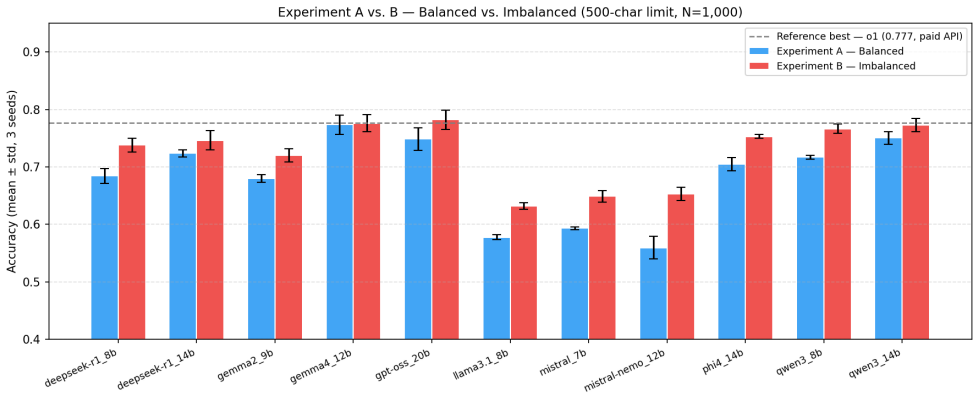


Fig. 2 Side-by-side comparison of accuracy (mean ± std, 3 seeds) across Experiment A (Balanced, blue) and Experiment B (Imbalanced, red) for all models (N=1,000). The dashed reference line marks the best published paid-API result (o1, 0.777). The upward shift in Experiment B partially reflects class imbalance rather than genuine model improvement.

overlapping but non-identical records, so this does not isolate a causal effect of class imbalance.

5.4 Comparison with Paid-API Models

Table 4 compares selected local results with published paid-API results from Roumeliotis et al. [8] (the complete plot is Supplementary Fig. S6) and summarises the key comparison points. The best archived zero-shot model, `gemma4:12b`, achieves a mean accuracy of 0.774 ± 0.017 and F1 of 0.774 ± 0.016 — numerically close to the paid GPT-4.5 (F1 0.7763) and within 0.003 of o1 (F1 0.7764) under the matched protocol, at zero API cost. Similar point estimates indicate competitive local performance but do not demonstrate statistical equivalence. `qwen3:14b` (F1 0.750) and `gpt-oss:20b` (F1 0.749) both have point estimates above the paid GPT-4o baseline (0.7358), although they remain below Claude 3.7 Sonnet (F1 0.7626). Because the reference paper does not publish the exact sampled complaint identifiers, this comparison should be read as protocol-matched rather than paired on identical instances. The cost differential is stark: the reference paper reported o1 costs of $24.24 per thousand predictions, versus $0.00 for all our models. Thus, Table 4 supports a strong result: the best local model (`gemma4:12b`) is numerically close to frontier paid models on this protocol while eliminating API fees and data transfer to third parties.

5.5 Narrative Length Sensitivity

Supplementary Fig. S8 compares mean accuracy on 250-character-limit versus 500-character-limit narratives for the balanced condition (both at N=1,000). The overall ranking is highly stable: no model changes its relative rank by more than one position between the two length limits. Most models improve by 0–3 percentage points moving from shorter to longer narratives, with `deepseek-r1:14b` (+1.6 pp) and `llama3.1:8b` (+2.8 pp) showing gains when additional text from the same complaints is retained.

Table 4 Zero-Shot Performance vs. API Cost (Balanced, N=1,000). Commercial API figures are from [8].

Model	Source	Accuracy	F1	Cost/1k
o1	Paid API	0.777	0.7764	$24.24
GPT-4.5	Paid API	0.774	0.7763	$12.32
`gemma4:12b`	**Open-source**	**0.774±.017**	**0.774±.016**	**$0.00**
Claude 3.7 Sonnet	Paid API	0.765	0.7626	$0.64
Gemini 2.0 Flash	Paid API	0.758	0.7586	$0.04
`qwen3:14b`	Open-source	0.751±.011	0.750±.011	$0.00
`gpt-oss:20b`	Open-source	0.748±.020	0.749±.019	$0.00
GPT-4o	Paid API	0.736	0.7358	$0.45
`deepseek-r1:14b`	Open-source	0.724±.006	0.724±.007	$0.00
`qwen3:8b`	Open-source	0.717±.003	0.717±.002	$0.00
`phi4:14b`	Open-source	0.705±.012	0.706±.011	$0.00

Pairing all 3,000 balanced predictions and applying Holm correction over 22 model–condition tests gives significant accuracy changes for `llama3.1:8b` ($p_{\text{adj}} < 0.001$) and `qwen3:14b` ($p_{\text{adj}} = 0.0234$). The DeepSeek-R1 14B change does not pass this corrected threshold ($p_{\text{adj}} = 0.0879$). `gpt-oss:20b` and `mistral-nemo:12b` show virtually no change (approximately 0.13 pp and 0 pp, respectively), indicating little observed sensitivity to narrative length in this range. The stability of the top-cluster ranking suggests that the ordering is not driven solely by these two length settings; this does not establish robustness to longer narratives.

5.6 Latency Analysis

Mean per-prediction latency is shown in the Lat.(s) column of Tables 2 and 3. Standard models are notably faster: `mistral:7b` completes a prediction in 0.23 s on average, while `phi4:14b` requires only 0.54 s despite being twice the parameter count. Reasoning models are slower: `deepseek-r1:14b` takes 9.33 s per prediction due to its end-to-end generation behaviour. `gemma4:12b` has the highest latency at 17.1 s. Separate thinking output, token counts, and termination reasons were not saved, so this latency cannot be attributed to a specific generation mechanism. The latency–accuracy trade-off is model-type dependent: `phi4:14b` achieves the best latency-accuracy ratio among all models (0.54 s, F1 0.706). The full Experiment B F1 plot is supplied as Supplementary Fig. S7.

5.7 Per-Class Breakdown for Leading Models

Table 5 reports per-class F1 for the three leading models in each experiment, confirming that aggregate weighted F1 masks important minority-class variation. In Experiment B all three models jump to $\geq$0.84 F1 on `credit_reporting` while minority classes remain in the 0.65–0.77 range, confirming the majority-class bias effect discussed in Section 6.

Table 5 Per-Class F1 for Leading Models (Exp. A and B, N=1,000)

E	Model	Retail	CreditRep.	Card	Mortg.	Debt
A	`gemma4:12b`	0.8247	0.7432	0.7634	0.8194	0.7178
A	`qwen3:14b`	0.7855	0.7212	0.7296	0.8097	0.7019
A	`gpt-oss:20b`	0.7939	0.7137	0.7394	0.7953	0.7045
B	`gpt-oss:20b`	0.7508	0.8527	0.6657	0.7404	0.6919
B	`gemma4:12b`	0.7676	0.8453	0.6719	0.7442	0.6769
B	`qwen3:14b`	0.7353	0.8437	0.6544	0.7427	0.6727

E = Experiment; Mortg. = Mortgages & Loans.

Table 6 Pooled balanced weighted F1 with 95 % cluster-bootstrap confidence intervals ($N = 3{,}000$).

Model	Pooled F1	95 % interval
`gemma4:12b`	0.7738	[0.7586, 0.7882]
`qwen3:14b`	0.7498	[0.7344, 0.7650]
`gpt-oss:20b`	0.7497	[0.7344, 0.7652]
`phi4:14b`	0.7057	[0.6892, 0.7216]

5.8 Additional Statistical and Output Audit

We estimate 95 % percentile confidence intervals using 5,000 bootstrap resamples of the pooled 3,000 predictions in each 500-character condition. Resampling is stratified by true class and clusters identical normalised narratives, so repeated texts receive the same bootstrap weight. These intervals condition on the saved test sets and model outputs, and do not include uncertainty about pre-training, prompt choice, or commercial reference results. The pooled estimators differ slightly from means of subset-specific F1.

The published GPT-4o point estimate (0.7358) falls within the local Qwen3 14B and gpt-oss intervals. This observation does not constitute a paired comparison or equivalence test, because reference predictions and sampling uncertainty are unavailable.

To examine class-prior effects, we rescale each row of the pooled Experiment B confusion matrix to equal support, retaining invalid predictions as an additional error column, then recompute metrics. For `mistral-nemo:12b`, observed accuracy 0.6530 becomes 0.5562, close to Experiment A's 0.5593. For `gpt-oss:20b`, accuracy 0.7820 becomes 0.7480, close to Experiment A's 0.7483. The weighted F1 values similarly change from 0.6413 to 0.5326 and from 0.7825 to 0.7495, respectively. Thus most of these aggregate gains can be explained arithmetically by different class proportions; this is not a causal experiment with the class prior independently manipulated.

Reparsing all 132 saved response files with a JSON decoder, accepting only an explicit string-valued `product` field, changes no in-taxonomy prediction under the retained legacy outputs. Across all Gemma 4 files, 37 saved responses are empty; their causes cannot be distinguished without separate thinking and termination logs. Representative Llama parse errors are explicit refusals, whereas Mistral 7B often returns parseable labels such as `online_banking` or `not_applicable` outside the taxonomy. These are distinct deployment failures even though both count as errors.

5.9 Supervised Baselines on the Preserved Test Sets

To assess whether an LLM is warranted when labelled data are available, we add TF-IDF/LinearSVC and fine-tuned DistilBERT [34] baselines. All twelve saved test sets remain unchanged. Training and validation use separate balanced sets of 20,000 and 2,000 records, respectively. We exclude normalised exact matches and shared 250-character prefixes with any test narrative, discard 133 conflicting-label text groups, and screen near duplicates using 128-permutation MinHash retrieval followed by exact word-trigram Jaccard similarity at threshold 0.85. This excludes 496 additional test-similar records and 28 validation-similar training candidates. The approximate retrieval cannot guarantee removal of every paraphrase.

Nested balanced training subsets of 500, 2,000, 10,000, and 20,000 records are used for the classical learning curve. TF-IDF uses word unigrams/bigrams, sublinear term frequencies, minimum document frequency 2, and at most 100,000 features. LinearSVC's C is selected from $\{0.1, 1, 10\}$ by validation weighted F1. Its pooled balanced-test weighted F1 increases from 0.7546 with 500 examples to 0.7908, 0.8241, and 0.8291 at the larger sizes. Warm serial CPU vectorisation and prediction take approximately 0.29 ms per narrative at 20,000 examples.

DistilBERT is trained for three epochs with learning rate 2×10^{-5}, batch size 16, weight decay 0.01, a 256-token cap, and 10 % warmup. The checkpoint with highest validation weighted F1 is selected. Training seeds are distinct from the three saved test-subset identifiers. Test performance is never used for checkpoint or hyperparameter selection. The training and prediction artifacts report exact model commits, training time, peak allocated VRAM, and warm serial GPU latency. Across three training seeds, mean pooled balanced weighted F1 is 0.8244 ± 0.0046 (standard deviation over model fits, each evaluated on the same 3,000 test records). Mean reduced-imbalance weighted F1 is 0.8220 ± 0.0042. Each fit takes approximately 80 s and peaks at 1.30 GiB allocated GPU memory; warm serial tokenisation and forward inference average 2.07 ms on the RTX 5060 Ti.

Paired two-sided McNemar comparisons against the saved Gemma 4 balanced predictions give Holm-adjusted $p < 0.001$ for the 20,000-record SVM and for each DistilBERT fit, correcting over all 42 baseline–reference–condition comparisons. This tests accuracy gains, not F1 differences; the zero-shot and supervised label budgets differ.

These supervised comparisons use additional labelled data and therefore measure a different deployment regime from zero-shot inference. Their strong performance requires the LLM deployment recommendations to be conditional on the availability of training labels. The matched 500-training-label LLM adaptation experiment is

Table 7 Supervised balanced-test performance. F1 is computed on the pooled 3,000 records; DistilBERT values average three separate model fits. Latency is warm serial end-to-end feature/token preparation and inference.

Model	Training records	Weighted F1	Latency (ms)
TF-IDF/SVM (CPU)	500	0.7546	0.29
TF-IDF/SVM (CPU)	2,000	0.7908	0.29
TF-IDF/SVM (CPU)	10,000	0.8241	0.28
TF-IDF/SVM (CPU)	20,000	0.8291	0.29
DistilBERT (GPU)	20,000	0.8244	2.07

Table 8 Paired HF base/QLoRA weighted F1 on 500-character tests. Each mode uses 3,000 pooled records; all adapters use 500 training and 2,000 validation records. A: balanced; B: reduced imbalance. Gains and paired 95 % intervals are in percentage points.

Model	Condition	Base F1	QLoRA F1	Δ (pp)	95 % CI (Δ)
Gemma 4	A	0.7520	0.7803	+2.83	[1.75, 3.96]
Gemma 4	B	0.7730	0.7948	+2.18	[1.12, 3.22]
Qwen3	A	0.7099	0.7891	+7.92	[6.55, 9.34]
Qwen3	B	0.7278	0.7946	+6.68	[5.30, 8.07]
Phi-4	A	0.6883	0.7845	+9.62	[8.21, 11.07]
Phi-4	B	0.7456	0.7994	+5.38	[4.08, 6.68]

reported next; higher-budget baselines do not establish equal-budget superiority. All supervised budgets additionally use the same 2,000 validation labels.

5.10 QLoRA Adaptation with 500 Training Records

All three models and both modes completed every preserved test file. Source/label alignment, original test and adapter hashes, raw-response reparsing, paired batch sizes and recomputed pooled metrics were verified for all 72,000 predictions. Table 8 and Fig. 3 report the 500-character comparisons. Balanced weighted F1 increases by 2.83 percentage points for Gemma, 7.92 for Qwen and 9.62 for Phi. The corresponding paired 95 % intervals are [1.75, 3.96], [6.55, 9.34] and [8.21, 11.07] percentage points. Qwen has the highest adapted balanced point estimate (0.7891), followed by Phi (0.7845) and Gemma (0.7803). These point estimates do not establish significant differences between adapted models.

All twelve paired F1 intervals exclude zero, including both length variants and reduced imbalance. All twelve accuracy comparisons pass Holm correction ($p_{\text{adj}} \leq 0.00142$). For 250-character balanced inputs, adapted F1 is 0.7674, 0.7767 and 0.7794 for Gemma, Qwen and Phi; gains are 2.13, 7.17 and 9.70 percentage points. Thus the gain is observed under both text limits, while their paired records preclude treating them as independent replications. Complete length, macro-F1 and class-specific results are supplied in the supplementary material.

Table 9 Local resource measurements for 500-training-record adapters. Latency: warm batch-1 mean on the same 100 balanced 500-character records per mode, including prompt preparation and generation. Training time includes validation/checkpointing and excludes initial weight loading. Memory is peak allocated CUDA memory, not total GPU utilisation.

Model	Base (s)	QLoRA (s)	Train (min)	Peak (GiB)
Gemma 4	0.475	0.647	34.75	7.77
Qwen3	0.421	0.539	41.15	10.04
Phi-4	0.487	0.435	40.68	9.32

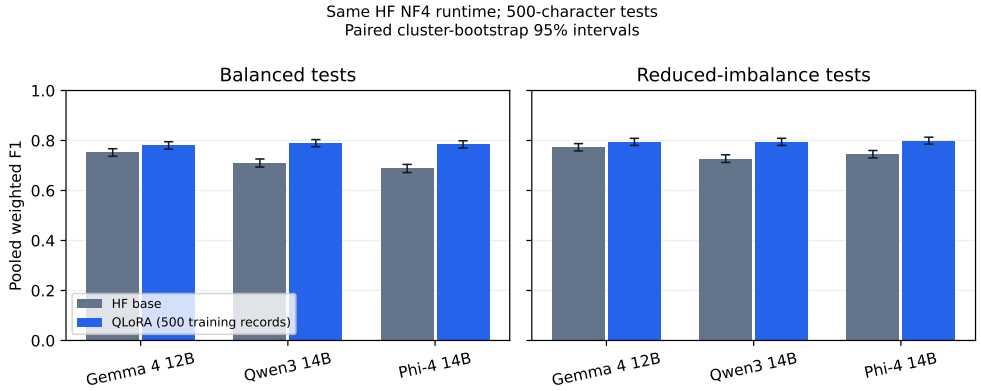


Fig. 3 Matched HF base/adapted pooled weighted F1 on 500-character tests, with 95 % cluster-bootstrap intervals. Each bar uses 3,000 records; adapters use 500 training and 2,000 validation records. The archived Ollama runtime is not used as the adaptation control.

In reduced imbalance at 500 characters, adapted weighted F1 is 0.7948, 0.7946 and 0.7994, whereas macro-F1 is 0.7576, 0.7602 and 0.7643. The gap retains the importance of minority-class evaluation. On balanced 500-character tests all five class F1 scores improve for each model; Qwen's largest class gain is debt collection (16.68 percentage points), and Phi's is retail banking (14.82 points). Across all test files, none of the 36,000 adapted predictions is invalid. Base Gemma has two invalid predictions, Qwen none and Phi 21. Improvements therefore extend beyond output-format repair: balanced Gemma and Qwen bases already have zero invalid predictions.

The training and serial timing results in Table 9 fit the 16 GB GPU budget. Adaptation does not consistently accelerate generation: Gemma and Qwen's measured serial means rise from 0.475 to 0.647 s and from 0.421 to 0.539 s, while Phi's falls from 0.487 to 0.435 s. These observations concern unmerged adapters and the measured 100-record sample, rather than a general serving-engine speedup. Serial/batch prediction agreement is 100/99 % for base/adapted Gemma, 98/100 % for Qwen and 100/100 % for Phi. The three differences have matching source hashes and input token counts;

their computational cause was not isolated. Full-test scores use the saved batched predictions.

6 Discussion

6.1 Reasoning vs. Standard Models

`gemma4:12b` achieves the highest weighted F1 in Experiment A (0.774 ± 0.016), numerically exceeding `gpt-oss:20b` (F1 0.749). However, Gemma 4 supports thinking, and the saved runs did not record whether it was active. This benchmark therefore compares deployed model configurations rather than isolating the benefit of reasoning. A controlled thinking-on/off comparison on identical complaints is needed to support that causal claim.

At the 8B tier, `deepseek-r1:8b` and `qwen3:8b` generally outperform similarly sized `gemma2:9b`, `llama3.1:8b`, and `mistral:7b`. At 14B, `qwen3:14b` (F1 0.750) and `phi4:14b` (F1 0.706) have different point estimates. These comparisons also change architecture and training recipe and cannot isolate a reasoning effect [22, 24].

An unexpected finding is the underperformance of `mistral-nemo:12b` (F1 0.540 ± 0.025 in Experiment A, rank 11 of 11), which falls below both its smaller sibling `mistral:7b` (F1 0.587) and all other 12B-tier models. Its off-label rate is low (0.7 per run), indicating that the model does produce parseable JSON but assigns incorrect categories. This suggests that the Mistral Nemo architecture, while competitive in general-purpose tasks, does not transfer as effectively to this narrow structured classification setting.

6.2 Within-Family Performance Patterns

Within the Qwen3 line, 14B exceeds 8B by 0.033 F1 points; the corresponding DeepSeek-R1 gain is 0.041. Both are within-line scale comparisons. Gemma 4 exceeds Gemma 2 by 0.098, but changes generation, size, and potentially thinking mode. Mistral-Nemo 12B falls below Mistral 7B despite having more parameters. These observations establish performance differences rather than their architectural causes. Detailed comparisons are supplied in the supplementary material.

6.3 Output Reliability and Inference Latency

Invalid predictions count as errors in accuracy, recall, and F1. Llama 3.1 8B records a mean of 67 parse errors in Experiment A and 57 in B, often explicit refusals. Mistral 7B records 59 and 42 out-of-taxonomy labels, usually parseable JSON. Both failure modes remain substantial. They require different remediation: response format enforcement does not by itself prevent refusals or resolve ambiguous labels. Reasoning-enabled builds generally show fewer invalid outputs, but architecture and alignment also differ, preventing a causal attribution to reasoning.

Among saved zero-shot runs, Phi-4 14B combines weighted F1 0.706 with 0.54 s mean latency; Mistral 7B is faster at 0.23 s but much less accurate. DeepSeek-R1 14B takes 9.33 s. Recorded latency reflects end-to-end server requests, rather than pure decoding throughput; separate thinking, token counts, and termination reasons were

not retained. The supervised baselines use direct classifier inference and their new serial latency measurements include feature/token preparation.

6.4 Deployment Recommendations

Deployment choices depend on available labels and the serving runtime. With 500 training labels and the common validation set, all three HF adapters improve their own matched base models, while TF-IDF/SVM provides a lower-latency option with pooled balanced F1 0.7546. At 20,000 training labels, SVM and DistilBERT have higher point estimates than the 500-label adapters and much lower measured serial latency. This is a comparison of deployment regimes, not equal-budget evidence that either model family is universally superior.

For local LLM adaptation, Phi-4 has the largest balanced F1 gain and the lowest adapted serial mean latency (0.435 s), while Qwen3 has the highest balanced adapted point estimate (0.7891). Their small F1 difference was not tested as a between-model superiority claim. Unmerged adapters increase measured serial latency for Gemma and Qwen; merging, alternative serving engines and concurrent request throughput were not evaluated. The following archived Ollama recommendations apply to their saved zero-shot configurations; they are not timing predictions for HF NF4.

Maximum accuracy (offline / batch): Use `gemma4:12b`. It achieves the highest balanced F1 (0.774), is numerically close to published paid-API scores, and has a reported artifact size of 7.6 GB. At 17.1 s per prediction, one million serial predictions require approximately 198 GPU-days on the measured workstation. Large archive processing would require independently measured batching or multiple GPUs; their speedup was not evaluated.

Real-time deployment (<1 s latency required): Use `phi4:14b`. It offers F1 0.706 at just 0.54 s per prediction — the best accuracy-to-latency ratio among all eleven models. At roughly 110 predictions per minute throughput, it can serve a moderate complaint routing queue in real time on the local GPU.

Smaller model artifacts: Consider `qwen3:8b` (5.2 GB artifact, F1 0.717, 5.19 s); peak runtime VRAM must be checked separately. It produces zero off-label predictions and outperforms all other sub-6 GB models.

Avoid `mistral:7b` and `llama3.1:8b` in zero-shot settings: both produce >50 off-label predictions per 1,000 samples on average, making them operationally unreliable without post-processing. `mistral-nemo:12b` is also not recommended for this task despite its size advantage.

Taken together, these recommendations translate into a concrete cost advantage. At moderate scale (one million complaints per year), deploying `gemma4:12b` instead of the best-performing paid API model (o1) would avoid approximately $24,240 in API fees at the historical reference rate. This is an API-fee comparison, not a total cost-of-ownership estimate or proof of equal quality. `qwen3:14b` and `gpt-oss:20b` similarly have point estimates above the published GPT-4o baseline (F1 0.7358) without API fees. This should not be read as computation being free: local inference still consumes GPU time, electricity, and local hardware resources. The practical gain is the elimination of recurring third-party API charges and the need to transfer sensitive complaint text to external services.

Equal-prior reweighting in Section 5.8 shows that most of the observed Experiment B gains reflect class proportions. Model quality should be assessed with macro-F1 and class-specific metrics; partly shared test records preclude treating A and B as independent replications.

6.5 Limitations

This study has several limitations. First, the balanced comparison matches the reference protocol but not necessarily the exact sampled instances, since the original sample identifiers are unavailable. Second, the 500-character length filter reduces inference cost but also shifts the class distribution, especially increasing the share of `credit_reporting`; future work should test whether the same patterns hold for longer narratives. Third, lowercasing and special- character removal may discard useful cues such as account-number patterns, capitalization, dates, or punctuation; the effect of less aggressive LLM-oriented preprocessing remains untested. Fourth, all experiments use N=1,000 examples per seed, evaluated at two narrative character-length limits (250 and 500 characters) across a five-class taxonomy; larger sample sizes, longer narratives, and richer taxonomies may reveal additional drift and rarer category confusions. Fifth, all local models are evaluated with a single zero-shot prompt. Prompt wording, output-format instructions, few-shot examples, or self-consistency decoding may change the relative ranking of models, particularly for smaller standard models with weaker instruction following. Sixth, the zero-shot models use compressed Ollama builds, including differing quantisation formats, so scores conflate model family, model size, and quantisation effects. Seventh, we cannot rule out pre-training contamination: open-weight models may have encountered CFPB complaints or derivative datasets during pre-training, although no task-specific fine-tuning is used in the LLM zero-shot runs. Eighth, LLM adaptation uses only 500 training records and one training seed. Its three test subsets and bootstrap intervals do not measure variation across independent adapter fits; larger training budgets and additional training seeds remain unevaluated for LLMs. The common validation set adds 2,000 labelled records to each reported supervised training budget. Serial latency is measured on 100 deterministic, class-balanced examples per mode after warmup, excluding response decoding and network serving overhead. Batch and serial outputs differ on three of 600 benchmark predictions; the full-test F1 estimates describe the saved batched outputs. Ninth, the reported $0 cost refers to API expenditure, not to the full economic cost of GPU hardware, electricity, or local workstation operation. Finally, the main tables report weighted aggregate metrics; for operational deployment, macro-F1 and per-class precision/recall should be inspected to avoid masking poor minority-class performance.

7 Conclusion

We compared archived zero-shot LLM configurations, controlled QLoRA adaptation and direct supervised classifiers on preserved CFPB tests. Using 500 balanced training labels and a common 2,000-label validation set, Gemma 4, Qwen3 and Phi-4 improve pooled balanced weighted F1 by 2.83, 7.92 and 9.62 percentage points under matched HF inference. Improvements remain positive in all twelve model–condition–length

comparisons, with no invalid adapted output. Training fits a 16 GB GPU and takes approximately 35–41 minutes. Adapted serial mean latency remains below 0.65 s in the measured benchmark, but adaptation does not uniformly accelerate inference.

The 20,000-training-label SVM and DistilBERT baselines achieve higher F1 point estimates with much lower classifier latency, so available labels and serving constraints determine which local approach is useful. Weighted scores under reduced imbalance must be interpreted alongside macro-F1 and class-specific results. The historical Ollama/API comparison provides zero-shot context, without establishing statistical equivalence or isolating reasoning effects. Future work should repeat LLM adaptation across training seeds and larger label budgets, evaluate longer narratives and temporal shifts, and measure alternative serving configurations.

Declarations

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Conflict of interest. The authors declare that they have no conflict of interest.

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The study uses the publicly available CFPB Consumer Complaints dataset published on Kaggle [28].

Materials availability. Not applicable.

Code availability. The original benchmark repository is available at: https://github.com/muhammettan28/ConsumerComplaintwithLLMs The adaptation pipeline records pinned model revisions, split manifests, source/adapter hashes and generated run artifacts for reproducibility.

Author contributions. Muhammet Tan and Halit Bakır contributed to the study conception, experimental design, implementation, analysis, and manuscript preparation.

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